

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) A manufacturer claims a population mean accuracy of 0.95. Which null hypothesis tests this claim? (1)
(A) $\mu = 0.95$
(B) $\bar{x} = 0.95$
(C) $\mu \neq 0.95$
(D) $\mu < 0.95$
- 2) A population has known standard deviation 0.02. For an independent sample of size 36, what is the standard error of the sample mean? (1)
(A) 0.12
(B) 0.02
(C) $0.02/6$
(D) $0.02/36$
- 3) With $\bar{x} = 0.947$, $\mu_0 = 0.95$, $\sigma = 0.02$ and $n = 36$, what is the z statistic? (1)
(A) 0.90
(B) -0.90
(C) -9.00
(D) 9.00
- 4) Observations range from 180 to 480. What is the width of each of three equal-width bins? (1)
(A) 60
(B) 150
(C) 300
(D) 100
- 5) Twelve observations are divided into three equal-frequency bins. How many observations are in each bin? (1)
(A) 3
(B) 4
(C) 6
(D) 12

Section B: Theory questions (25 marks)

- 6) (i) Find the standard deviation and skewness for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Give an example of the combining-variables concept in dataset preparation.
(iii) A CMO compares three new cola flavours using ratings from 12 customers on a 1-9 scale. Conduct the appropriate analysis. Ratings: Flavour A: 4, 2, 4, 2; Flavour B: 6, 4, 5, 5; Flavour C: 8, 6, 8, 6. Use population standard deviation and Pearson's second skewness coefficient. For (iii), assume independent normal samples with equal variances; use a one-way ANOVA at 5%, with critical $F(2,9) = 4.256$.
- 7) (i) Find the elements of a box plot for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Why is it sometimes necessary to create subsets from a large dataset before analysis?
(iii) Measure the strength of the relationship between the rankings assigned by two interviewers to candidates A-L. Interviewer 1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. Interviewer 2: 1, 2, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11. For quartiles, use medians of the two halves. Use Spearman rank correlation in (iii).
- 8) A hospital laboratory claims that a new COVID testing kit correctly identifies infection 95% of the time. A researcher believes the true accuracy is less than 95%. A sample of 50 test results has mean accuracy 0.93; the known population standard deviation is 0.04. Test the claim at the 5% significance level. Critical z-value: -1.645. (3)
- 9) Test the effectiveness of vaccination in preventing smallpox at the 5% significance level. (5)
Vaccinated: attacked 31, not attacked 469, total 500.
Not vaccinated: attacked 185, not attacked 1315, total 1500.
Overall: attacked 216, not attacked 1784, total 2000. Use Pearson chi-square without continuity correction; critical value for 1 degree of freedom is 3.841. State association, not a causal conclusion.
- 10) Per-acre wheat production (tons) is recorded for three varieties and four fertilizers, with each combination replicated twice. Test whether variety differences and fertilizer differences are significant at the 5% level. Each pair below gives the two replicates, in fertilizer order F1, F2, F3, F4. (5)
Variety 1: (8,9), (9,8), (6,7), (10,11).
Variety 2: (7,6), (8,9), (5,4), (9,10).
Variety 3: (10,9), (11,12), (8,7), (12,13). Assume independent normal errors with common variance. Include the interaction term. Use critical F values: variety (2,12): 3.885; fertilizer (3,12): 3.490.
- 11) A data scientist is analysing the following cholesterol level (mg/dL) readings of 12 patients: [145, 155, 160, 162, 165, 172, 180, 190, 200, 220, 240, 260]. Apply equal-width and equal-frequency binning with 3 bins. Using histogram analysis, explain which binning approach (width-based or frequency-based) better represents the data distribution and why. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA STRUCTURES [CSS 2101]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2; module 3 restricted to stacks. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which data structure directly supports undoing the most recent action first? (1)
(A) Queue
(B) Stack
(C) Sorted array
(D) Adjacency matrix
- 2) Edits A, B and C are pushed in that order onto an undo stack. Which edit does the first undo remove? (1)
(A) A
(B) B
(C) All three
(D) C
- 3) After edits A, B and C, two undo operations transfer actions to a redo stack. What is the redo stack from bottom to top? (1)
(A) C, B
(B) B, C
(C) A, B
(D) A, C
- 4) The redo stack contains C, B from bottom to top. Which edit is reapplied by one redo operation? (1)
(A) A
(B) C
(C) B
(D) Both B and C
- 5) In a conventional two-stack editor, what happens to the redo stack when a new edit is made after an undo? (1)
(A) It is reversed
(B) It is cleared
(C) It is unchanged
(D) It becomes the undo stack

Section B: Theory questions (25 marks)

- 6) Write a C program to read and display employee details including the date of joining using nested structure. Consider date of joining as nested structure. (2)
- 7) Assume that you are storing marks of students in 3 subjects. Write a C program that: i. Takes the number of students as input. ii. Dynamically allocates a 2D array using pointers. iii. Accepts marks for each subject. Calculates and prints the average marks of each student using only pointer arithmetic. (2)
- 8) Write a C program to dynamically allocate memory for 2D string arrays (array of strings) and perform concatenation using pointers. (2)
- 9) A music streaming service keeps track of songs in a playlist using a singly linked list. Each node stores the following details: SongID (integer), Title (string), Duration (in seconds). Write a C program to perform the following operations: Insert songs into the playlist in the order they are added. Remove all songs whose duration is less than a user-specified value (Use head pointer for performing this operation). Display the updated playlist. Include a main() function to demonstrate all functionalities. (5)
- 10) A mobile app named "MIT-BLR Connect" maintains contact details of alumni in three different modules: i) Add Recent Contacts Module: Displays recently interacted alumni in order of contact - new ones added at the beginning. ii) Traverse Module: Allows viewing alumni details in forward direction. iii) Develop a module to print the data in reverse order. Develop C functions considering singly linked list as the suitable data structure (5)
- 11) Write a function to convert an infix expression to postfix. Include the following functions as per the prototype. i) int InStackPrecedence(char op) ; to return the precedence of operator within the stack ii) int IncomingPrecedence(char op); to return the precedence of the incoming operator. iii) void Infix_Postfix(char infix[], char postfix[]); to convert given infix expression to postfix. Main function not required. Note: The infix expression may include ONLY the following operators (listed in the order of their precedence): ; ^ (exponentiation) → right associative ; * (multiplication), / (division) → left associative ; + (addition), - (subtraction) → left associative (5)
- 12) Write a C function bool isDuplicate(char s[]) that detects duplicate parentheses in a given expression using a stack. (2)
- 13) Using the following nested calls, justify why a stack is preferred over a queue in handling function calls. Explain the role of activation records (stack frames).
void fun1() { printf("Inside fun1\n"); }
void fun2() { fun1(); printf("Inside fun2\n"); }
int main() { fun2(); printf("Inside main\n"); } (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA COMMUNICATION AND COMPUTER NETWORKS [CSS 2102]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) A 900-byte payload passes through five layers, each adding a 20-byte header. With no trailers, what is the transmitted size? (1)
(A) 920 bytes
(B) 1000 bytes
(C) 1020 bytes
(D) 1100 bytes
- 2) A transmitted unit contains 900 payload bytes and 100 header bytes. What percentage of the total is headers? (1)
(A) 9%
(B) 11.11%
(C) 90%
(D) 10%
- 3) What is the polynomial degree of CRC generator 10011? (1)
(A) 3
(B) 5
(C) 4
(D) 2
- 4) How many zeros are appended to the data before CRC division using generator 10011? (1)
(A) 4
(B) 5
(C) 3
(D) 1
- 5) Which operation performs subtraction in CRC modulo-2 division? (1)
(A) AND
(B) XOR
(C) OR
(D) Decimal subtraction

Section B: Theory questions (25 marks)

- 6) Compare Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) in terms of how each modifies the carrier signal, their bandwidth efficiency, and susceptibility to noise. Analyse which technique is most suitable for transmission over a noisy telephone channel and justify your choice. (6)

- 7) Define data rate and bandwidth. A square wave is approximated by (6)

$$s(t) = \frac{4}{\pi} \left[\sin(2\pi \cdot 2 \cdot 10^6 t) + \frac{1}{3} \sin(2\pi \cdot 6 \cdot 10^6 t) + \frac{1}{5} \sin(2\pi \cdot 10 \cdot 10^6 t) \right].$$

Assume two bits per fundamental cycle and bandwidth equal to highest minus lowest retained frequency. Calculate the data rate and bandwidth. Triple all frequencies and find the new values. Explain how bandwidth affects data rate.

- 8) For the given binary data 101000101010, draw the following line coding scheme. (Assume previous pulse was negative) i. Differential Manchester ii. Bipolar AMI For Differential Manchester, use a boundary transition for 0 and none for 1; initial level is negative. A mid-bit transition always occurs. (2)
- 9) For the given binary data 111000101010, draw the following line coding scheme. (Assume previous pulse was negative) i) Manchester ii) Bipolar AMI Manchester convention: 1 is low-to-high and 0 is high-to-low at mid-bit. (2)
- 10) A company network consists of three departments connected as follows: Sales and HR are connected through a hub, HR and IT are connected through a switch, and the switch is connected to a router for internet access. An employee in Sales sends a file to an employee in IT. Analyse how the hub, switch, and router handle this transmission. (2)
- 11) A sender uses a window of 3 to transmit frames 0, 1 and 2. Frame 1 is lost, while 0 and 2 arrive. ACKs are immediate and never lost; no new frames are available. In go-back-N, discard out-of-order frames and retransmit all outstanding frames at timeout. In selective repeat, buffer out-of-order frames and retransmit only the unacknowledged frame. For each protocol, list retransmissions and calculate useful frames divided by total data transmissions. (3)
- 12) Calculate the total time required to transmit 1,000,000 bits using the Stop-and-Wait ARQ protocol when each frame carries 2000 bits, the sender and receiver are 5000 km apart, and the propagation speed is 2×10^8 m/s. Ignore transmission, waiting, and processing delays, and assume no frames are lost or damaged Count one full round-trip per frame, including the final ACK. (2)
- 13) An Ethernet segment has transmission speed 10^8 bits/second and maximum length 1500 metres. If propagation speed is 3×10^8 metres/second, calculate the minimum frame size in bits required for collision detection. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) What is the range of 5-bit two's-complement integers? (1)
(A) -15 to 15
(B) -16 to 15
(C) -16 to 16
(D) 0 to 31
- 2) What is the 5-bit two's-complement representation of -9 ? (1)
(A) 01001
(B) 11001
(C) 10110
(D) 10111
- 3) Which addition overflows the 5-bit two's-complement range? (1)
(A) $-9 + 5$
(B) $6 - 13$
(C) $11 + 7$
(D) $-4 + 3$
- 4) Which arrays are programmable in a programmable logic array (PLA)? (1)
(A) Both AND and OR
(B) Only AND
(C) Only OR
(D) Neither
- 5) A PLA implements $F_1 = AB + AC$ and $F_2 = AB + BC$. Which product term can be shared? (1)
(A) AC
(B) AB
(C) BC
(D) ABC

Section B: Theory questions (25 marks)

- 6) Simplify $F(A, B, C, D) = \Pi M(0, 3, 5, 6, 8, 10)$, with don't-care terms $d(2, 13, 14)$, and implement it using NOR gates only. Take A as the most significant bit. (3)
- 7) Design a 3-bit magnitude comparator for unsigned binary numbers $A = a_2a_1a_0$ and $B = b_2b_1b_0$. Derive the Boolean expressions for $A > B$, $A = B$ and $A < B$, and draw the equivalent logic circuit. (3)
- 8) Implement a three-input XOR gate using 4-to-1 multiplexer with a neat diagram. (3)
- 9) Design a sequential circuit with two JK flip-flops A and B and two inputs E and F. If $E = 0$, the circuit remains in the same state regardless of the value of F. When $E = 1$ and $F = 1$, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When $E = 1$ and $F = 0$, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats. (5)
- 10) Design a 3-bit synchronous binary up-counter using T flip-flops. Draw the logic diagram. (3)
- 11) Illustrate a 4-bit asynchronous ripple counter with a neat diagram using JK Flip flops and explain the working of the same. (3)
- 12) Write behavioral Verilog code that represents an eight-bit ring counter. (2)
- 13) Explain the concept of memory addresses in an Instruction Set Architecture. Also, discuss the difference between byte addressing and word addressing. (3)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

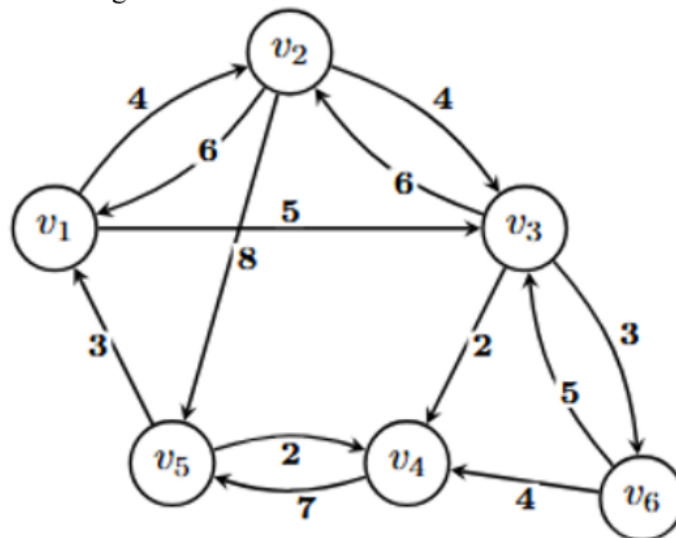
Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which pair of conditions defines a complement b of a in a Boolean algebra? (1)
(A) $a \vee b = 0, a \wedge b = 1$
(B) $a \vee b = 1, a \wedge b = 0$
(C) $a \vee b = a, a \wedge b = b$
(D) $a = b$
- 2) How many complements does each element of a Boolean algebra have? (1)
(A) None
(B) At least two
(C) Infinitely many
(D) Exactly one
- 3) Which expression equals $a \vee (\neg a \wedge b)$? (1)
(A) $a \vee b$
(B) $a \wedge b$
(C) $\neg a \vee b$
(D) a
- 4) Which expression is in disjunctive normal form (a disjunction of conjunctions of literals)? (1)
(A) $(x \vee y) \wedge z$
(B) $\neg(x \vee y)$
(C) $(x \wedge y) \vee (\neg x \wedge z)$
(D) $(x \vee y) \wedge (\neg x \vee z)$
- 5) Which expression is in conjunctive normal form (a conjunction of disjunctions of literals)? (1)
(A) $(x \wedge y) \vee z$
(B) $(x \vee y) \wedge (\neg x \vee z)$
(C) $\neg(x \wedge y)$
(D) $(x \wedge y) \vee (\neg x \wedge z)$

Section B: Theory questions (25 marks)

- 6) Draw the Hasse diagram for $A = \{1, 2, 3, 12, 18, 36\}$, ordered by divisibility: a is related to b iff a divides b . Is it a lattice? Justify. List all antichains of length more than one. (5)
- 7) How many ways are there to select a total of 15 items from three types, subject to these limits? (4)
Type A: 1 to 4 items; Type B: 3 to 7 items; Type C: 5 to 10 items.
- 8) Show that the number of partitions of n with no part smaller than 3 equals the number of partitions of $n+3$ whose three largest parts are consecutive integers. (3)
- 9) Find the 75th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order. (3)
- 10) Show that if G is a regular self-complementary graph of order n , then $n = 4t + 1$ for some integer t . (3)
- 11) If G has p vertices and minimum degree $\delta(G) \geq (p - 1)/2$, prove that G is connected. (3)
- 12) Use Dijkstra's algorithm to find the shortest paths from v_1 to every other vertex in the directed network below. Show the steps. (4)



End of question paper